

CMJ Force Plate Thresholding Framework

Athlete Classification & Training Intervention Decision Guide

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Purpose & Scope

This framework operationalizes countermovement jump (CMJ) force plate data collected via VALD ForceDecks into a standardized decision protocol for athlete classification, training group assignment, and targeted neuromuscular intervention. It is designed to bridge the gap between testing data and daily training prescriptions for basketball athletes throughout the competitive season.

The thresholding system is grounded in NBA, NBA G-League and NCAA Division I basketball normative data (Berberet et al., 2024; Philipp et al., 2023; VALD Health, 2025) and validated monitoring literature (Schuster et al., 2020; Claudino et al., 2017; Pentheny, 2025), adapted for the professional NBA and G League context. All primary thresholds are anchored directly to NBA-level normative percentile data (VALD Health, 2025), allowing practitioners the ability to interpret athlete values relative to the professional standards. Thresholds reflect position specific demands and are applied both at baseline (preseason classification) and longitudinally (weekly readiness monitoring).

Primary Threshold Metrics

The following six CMJ metrics drive classification and intervention decisions. NBA normative percentile values are provided for each metric. Gold Group thresholds are set at or above the P75; Purple Group spans P25–P75; White Group falls below P25. All values are from the VALD ForceDecks NBA normative dataset unless otherwise noted.

Metric	NBA Norms (P25 / P50 / P75)	GOLD Threshold(≥ P75)	PURPLE Range(P25–P75)	WHITE Flag(<P25)	Source
mRSI-Mod	0.52 / 0.60 / 0.69	≥0.69	0.52–0.69	<0.52	VALD,2025; Schuster et al., 2020; Philipp et al. 2023; Pentheny, 2025
Peak Power/BM (W/kg)	56.1 / 60.8 / 66.4	≥66 W/kg	56–66 W/kg	<56 W/kg	VALD,2025; Cabarkapa et al.,2023; Philipp et

Metric	NBA Norms (P25 / P50 / P75)	GOLD Threshold($\geq$ P75)	PURPLE Range(P25–P75)	WHITE Flag(<P25)	Source
					al.,2023;Pentheny , 2025
EDRFD/BM Eccentric Deceleration RFD (N/s/kg)	80 / 105 / 137	≥ 137 N/s/kg	80–137 N/s/kg	<80 N/s/kg	VALD NBA Norms; Harper et al.,2020;Pentheny , 2025
Eccentric Peak Power/BM (W/kg)	17.4 / 21.1 / 25.3	≥ 25 W/kg	17–25 W/kg	<17 W/kg	VALD, 2025; McMahon et al. 2018;Pentheny, 2025
Bilateral Asymmetry Braking Impulse (%)	NBA NordBord Asym P50=8%; CMJ asymmetry benchmarks vary	<10%(low risk)	10–15%(flag + PT)	>15%(restriction)	Heishman et al., 2019; Claudino et al.,2017;Pentheny , 2025
Countermovement Depth (cm, absolute)	P25=26.6 / P50=30.4 / P75=34.4(deeper = less efficient)	≤ 26.6 cm(efficient, shallow)	26.6–34.4 cm(average range)	>35 cm guards>40 cm bigs(SSC flag)	VALD, 2025; McMahon et al.,2018; Krzyszkowski et al.,2022;Pentheny , 2025

Note: NBA normative data = VALD ForceDecks NBA dataset (all positions combined). EDRFD = Eccentric Deceleration Rate of Force Development; directly quantifies the ability to generate braking force rapidly, the most sensitive CMJ metric for deceleration injury risk. CM Depth note: deeper depth (larger absolute value) = less SSC efficiency; shallower depth at equal jump height = superior SSC coupling. Individual baselines established during preseason take precedence over population norms for in-season monitoring decisions.

Secondary & Contextual Metrics - CMJ

The following metrics provide supporting context for classification decisions and are especially valuable when primary metrics present conflicting signals or when profiling athlete neuromuscular archetypes. All NBA normative values are from the VALD NBA Normative dataset.

Metric	NBA P25	NBA P50	NBA P75	Clinical Significance	Source
Jump Height Flight Time (cm)	38.2	42.8	47.0	Global output benchmark; used for percentile reporting and athlete communication	VALD (2025)
Concentric Impulse (N·s)	254	278	312	Total concentric output; primary	VALD (2025)

Metric	NBA P25	NBA P50	NBA P75	Clinical Significance	Source
				driver of P1/P2 quartile classification (see Section 4)	
Concentric Impulse 100ms (N·s)	136	160	186	Quantifies early-phase force production; direct objective proxy for P1 concentric capability; pairs with RPD 100ms for rate-specific profiling	VALD (2025)
Concentric RPD 100ms/BM (W/s/kg)	308	398	493	Rate of power development in the first 100ms; superior to peak power for identifying explosive deficit vs. capacity deficit; key discriminator of training response	VALD,2025; McMahon et al. 2018
Force at Zero Velocity FV0 (N/kg)	23.0	25.3	27.8	Eccentric-to-concentric transition force; indicator of tendon stiffness and SSC efficiency. Low FV0 relative to depth = tendon/SSC compliance concern	VALD, 2025; McMahon et al.,2018
Lower-Limb Stiffness (N/m)	6,094	7,713	9,352	<P25 combined with high CM depth and low mRSI = tendon compliance + reactive deficit cluster; trigger for isometric/tendon protocol intensification	VALD,2025; Cabarkapa et al.,2023
Eccentric Peak Velocity (m/s, absolute)	1.20	1.35	1.49	How aggressively an athlete loads the eccentric phase; high velocity + low EPP/BM = aggressive but inefficient braking; low velocity + low mRSI = passive, slow loading pattern	VALD, 2025; McMahon et al., 2018
Eccentric Duration (s)	0.413	0.454	0.497	Prolonged eccentric (>P75) + low peak	VALD (2025)

Metric	NBA P25	NBA P50	NBA P75	Clinical Significance	Source
				velocity = slow, inefficient loading; short duration + low EDRFD = high-speed but shallow braking — poor shock absorption	
Peak Landing Force/BM (N/kg)	41	48	57	Deceleration/landing load tolerance; low landing force relative to jump height may indicate landing avoidance/guarding behavior — post-injury return-to-play marker	VALD, 2025; Hart & Chumanov, 2025.

Secondary & Contextual Metrics

The following metrics provide supporting context for classification decisions but are not used as primary sorting criteria. They are especially useful when primary metrics present conflicting signals or when profiling athlete neuromuscular archetypes.

Metric	Benchmark	Clinical Significance	Source
Braking Net Impulse (N·s)	≥143 N·s (50th %ile, forwards)	Low braking impulse = poor eccentric recruitment or delayed loading; correlates with deceleration deficits	Wagle et al. (2023); McMahon et al. (2018); Pentheny, (2025)
Time to Takeoff (s)	~0.55–0.65 s normative range	Prolonged TTTO (>0.70 s) + excessive depth = inefficient SSC coupling; post-injury marker	McMahon et al. (2018); Hart & Chumanov (2025); Pentheny, (2025)
Force at Zero Velocity (FvO, kg/bw)	Guards: ~1900–2200 N NBigs: ~2200–2600 N	Transition point between eccentric/concentric; indicator of tendon/SSC stiffness	McMahon et al. (2018); Cabarkapa et al. (2023); Pentheny, (2025)
Eccentric Mean Force (N)	Guards/Forwards ~800–1,000 N avg	<50th %ile signals inadequate eccentric capacity for basketball decel demands	Berberet et al. (2024); Harper et al. (2024); Pentheny, (2025)

P1/P2 Concentric Impulse Phase Analysis

Beyond total concentric impulse, the distribution of force production within the concentric phase provides critical information for individualized programming. P1 represents the impulse generated in the first 50% of the concentric phase (initial drive); P2 represents the impulse generated in the final 50% (finishing/completion) (VALD, 2025). The ratio and absolute values of each phase reveal where in the jump an athlete under-produces and therefore where training emphasis should be directed (Pentheny, 2025; Claudino et al., 2017; McMahon et al., 2018)

This framework was developed through the Washington Nationals professional baseball CMJ dataset and has been adapted here for NBA and NBA G-League basketball applications. Athletes are first sorted into four quartiles by total concentric impulse, then each quartile is subdivided into four quadrants based on the P1/P2 split. This creates an individualized gradient for programming decisions within a classification of training groups.

Concentric Impulse Quartile Benchmarks (P1/P2 Framework)

The quartile structure below was developed through longitudinal CMJ data from professional baseball (Washington Nationals, N=317) and adapted for basketball application. Quartile boundaries should be recalibrated annually to your roster's preseason distribution. NBA normative concentric impulse percentiles (P25=254, P50=278, P75=312 N·s) are provided as an external reference anchor.

Quartile	Conc. Impulse Range (N·s)	NBA Percentile (ref.)	Jump Height (FT) est.	Performance Context
Q1	~180–248 N·s	Below P25	~14–16 in	Below-average total concentric output; highest intervention priority; address total capacity before phase-specific targeting
Q2	~248–274 N·s	~P25–P50	~16–17 in	Below-average to average; highest-yield band for P1/P2 quadrant targeting — athletes have adequate total impulse, but phase-specific inefficiency drives the deficit
Q3	~274–295 N·s	~P50–P75	~17–18 in	Above-average; typically Gold Group athletes. P1/P2 ratio identifies residual phase-specific inefficiency limiting Q3→Q4 progression
Q4	~295–392 N·s	Above P75	>18 in	Elite output; within-group P1/P2 optimization is the primary lever — athletes benefit most from targeted phase training over general volume addition

P1/P2 framework developed from Washington Nationals professional baseball CMJ dataset; adapted for basketball (Pentheny, 2025). NBA Concentric Impulse 100ms (closest NBA normative P1 anchor): P25=136 N·s, P50=160 N·s, P75=186 N·s. Concentric RPD 100ms/BM: P25=308, P50=398, P75=493 W/s/kg (VALD 2025). Recalibrate quartile boundaries annually from your preseason roster distribution. For the 2026–27 NBA season, baseline data will be drawn from the 2025–26 NBA season data.

P1/P2 Quadrant Classification Within Each Quartile

Within each concentric impulse quartile, athletes are subdivided into four quadrants based on their individual P1 and P2 values relative to the quartile median. This reveals the specific phase of the concentric movement where an athlete produces disproportionately more or less force, and maps directly to targeted training interventions.

Quadrant	P1 / P2 Profile	Movement Interpretation	Training Emphasis	Example Methods
Low P1(High P2)	P1 below quartile median; P2 at or above median	Athlete struggles to generate force early in the push-off; compensates with a strong finish. Suggests slow initial RFD, poor early-phase explosive capacity.	Improve early-phase concentric RFD; enhance initial drive off the floor	Heavy trap bar DL (0.5–0.6 m/s zone); pause squat at parallel; isometric mid-thigh pulls; ballistic squat with intent cues
Low P2(High P1)	P1 at or above median; P2 below median	The athlete drives hard initially but loses force output in the finishing phase. Suggests top-end power deficit or inability to maintain RFD through full ROM.	Develop top-end power and full-ROM force application; finishing velocity	Jump squat (light load, 20–30% 1RM); loaded CMJ; altitude drop to max CMJ; Olympic lift variations (hang clean, push jerk) for triple-extension finish
Balanced	P1 and P2 both at or above quartile median	Proportionate force distribution across the full concentric phase. Athlete's total concentric impulse is the primary limiting factor, the profile is efficient, just needs volume.	General progressive overload; volume-driven concentric development	Standard VBT squat/DL progressions; CMJ volume; reactive bounding; focus is load tolerance and progressive overload, not phase-specific targeting
Low BW(Both Low)	Both P1 and P2 below quartile median; body mass may be a contributing factor	Global concentric force deficit relative to body weight. Athlete lacks capacity throughout the entire push-off phase. Most common in heavier athletes at lower quartiles.	Improve relative power output; body composition, if relevant; general strength base	General hypertrophy + strength base (full-range compound movements); body composition management; VBT at moderate loads (0.6–0.8 m/s); plyometric volume building

P1/P2 Integration with Training Group Classification

P1/P2 quadrant profiling operates as a layer of specificity within each training group — it does not replace Gold/Purple/White classification but refines the training emphasis within that group. The matrix below shows how quadrant type interacts with group assignment:

Quadrant	If Gold Group	If Purple Group	If White Group
Low P1	Add 1–2 early-phase exercises (pause squat, IMTP); maintain full Gold volume	Prioritize early-phase RFD work; deload top-end plyos until P1 improves	Isometric mid-thigh pull; pause holds at parallel; no plyometric work
Low P2	Emphasize loaded CMJ and jump squat at end-range; Olympic pull variations	Reduce maximal strength emphasis; increase light-load ballistic work	Bodyweight jump squats with focus on full triple-extension only (no reactive work)
Balanced	Standard Gold programming — progressive overload is the primary lever	Focus on total volume and mRSI recovery; no phase-specific targeting needed	Standard White structural loading; no phase-specific adjustments
Low BW	Investigate body composition; volume management; relative power metrics	General strength base before phase-specific work; prioritize relative strength ratio	Clinical + nutrition review if applicable; structural load only

P1 and P2 values are extracted directly from the VALD ForceDecks concentric phase report. The split point is defined as 50% of total concentric duration (time from minimum displacement to takeoff). Both absolute N·s values and the P1:P2 ratio should be tracked longitudinally — a narrowing ratio in a Low P1 athlete indicates positive early-phase RFD development.

CMJ Classification Decision Tree

The decision tree is applied at each formal testing point (preseason, monthly, and following clinical events). Athletes are sorted into three training groups based on CMJ threshold performance. Group assignment drives programming selection (see Gold/Purple/White Group document) and triggers appropriate intervention protocols.

STEP 1: Asymmetry Screen (Applied Before All Other Thresholds)

Bilateral asymmetry is assessed first as it represents an independent risk flag regardless of absolute performance levels.

- **PASS** Asymmetry <10%: Proceed to Step 2.
- **CAUTION** Asymmetry 10–15%: Flag for review; proceed to Step 2 with notation.
- **FLAG** Asymmetry >15–20%: Restrict bilateral high-load training; assign modified White Group programming; initiate clinical evaluation within 48 hours.

STEP 2: mRSI Screen (Primary Classification Driver)

mRSI is the single most sensitive marker of in-season neuromuscular readiness for basketball athletes (Schuster et al., 2020; Philipp et al., 2023). It is the primary sorting variable.

mRSI Value (Guards)	mRSI Value (Bigs)	Classification Signal	Initial Assignment
>0.80 m/s	>0.75 m/s	High neuromuscular readiness	→ GOLD Group (confirm w/ Step 3)
0.69–0.80 m/s	0.65–0.75 m/s	Adequate; proceed to Step 3 for confirmation	→ Conditional GOLD or PURPLE (Step 3 determines)
0.60–0.69 m/s	0.55–0.65 m/s	Below-threshold reactive strength; likely fatigue or structural concern	→ PURPLE Group (proceed to Step 3 for subgroup)
<0.60 m/s	<0.55 m/s	Significant deficit; restrict high-velocity/impact training	→ WHITE Group (clinical review protocol)

STEP 3: Eccentric Profile Confirmation (Differentiation Within Gold/Purple)

For athletes not clearly assigned in Step 2 (mRSI 0.65–0.80 for guards), eccentric peak force and peak power confirm or revise the group assignment.

- **GOLD CONFIRM:** EPF $>2.5 \times \text{BW}$ AND Peak Power $\geq 68 \text{ W/kg}$: Confirm Gold Group.
- **PURPLE ASSIGN:** EPF $2.0\text{--}2.5 \times \text{BW}$ OR Peak Power $63\text{--}68 \text{ W/kg}$: Assign Purple Group — eccentric-emphasis programming.
- **ECCENTRIC FLAG:** EPF $<2.0 \times \text{BW}$ regardless of mRSI: Assign Purple Group — structural eccentric deficit; flag for clinical review if bilateral.
- **DEPTH FLAG:** Countermovement depth $>30 \text{ cm}$ (guards) or $>38 \text{ cm}$ (bigs): Note SSC coupling deficit regardless of group; trigger depth-reduction protocol.

STEP 4: Longitudinal Monitoring (Weekly Readiness Protocol)

Following preseason classification, CMJ testing occurs 2 to 3 times per week (standard), with daily monitoring available during high-load periods. Longitudinal decisions use individualized baseline thresholds (established during preseason), NOT population norms.

- **YELLOW FLAG:** Drop of $>5\%$ from individual baseline in mRSI across 2 consecutive tests: Training load reduction $15\text{--}20\%$; monitor next 48 hours.
- **ORANGE FLAG:** Drop of $>10\%$ from baseline across 2+ tests: Shift to Purple Group programming for the current microcycle; clinical check-in.
- **RED FLAG:** Drop of $>15\%$ from baseline OR acute single-session drop $>10\%$: Restrict to White Group that day; same-day clinical triage.
- **ASYMMETRY ALERT:** Asymmetry increase $>5\text{--}10\%$ from individual baseline: notification regardless of absolute value.

Intervention Protocols by Metric Deficiency

When CMJ metrics fall below the threshold, the following targeted interventions are applied as adjuncts to or modifications of the athlete's group program. These protocols address specific deficits rather than replacing full training.

Deficit Identified	Primary Metric Flag	Intervention Focus	Example Methods
Low mRSI / SSC Inefficiency	mRSI <0.55 (guards)	Improve eccentric-to-concentric transition; reduce countermovement depth	Accentuated eccentric loading; drop landing progressions; continuous jump series; reactive drills with intent cues
Eccentric Deficit	EPF <2.0 × BW	Increase eccentric strength capacity; improve deceleration loading tolerance	Eccentric isometric split squat; Nordic hamstring; slow-tempo RDLs (5s eccentric); flywheel or 1080 loading
Bilateral Asymmetry	Asymmetry >12-15% impulse	Restore limb symmetry; address compensatory patterns; rule out structural cause	Unilateral loading emphasis (split squat, single-leg RDL, step-downs); weekly symmetry trend monitoring; PT-guided assessment
Low Peak Power	PP <50 W/kg (guards)	Build concentric power output; improve RFD in the propulsive phase	VBT-guided squats/trap bar DL (0.75–1.0 m/s targets); loaded CMJ; box jump progressions; potentiation complex methods
Tendon Stiffness / Reactive Deficit	Low FV0, high TTTO	Improve tendon mechanical properties; enhance SSC stiffness	Isometric holds at tendon angle; pogo hops; ankle/Achilles progressive loading; slow heavy calf (3-6s ISO + slow eccentric)

Assessment Cadence & Integration

Testing Schedule

- **Baseline:** Preseason (Aug/Sep): Full CMJ tests bilateral, unilateral, drop jump; establishes individual baselines.
- **In-Season:** At minimum 2–3× per week (pregame or highest intent output) bilateral akimbo CMJ only (2–3 jumps, best 2–3 averaged).
- **Monthly:** Monthly formal: Full battery re-test; group reclassification assessment.
- **Post-Game:** Post-game: Optional single CMJ set within 60 min of final buzzer for acute load snapshot when RTP or other unique case scenario.
- **RTP:** Post-injury: Full battery before return to unrestricted training.

Reclassification Criteria

Athletes may be reclassified between groups at any formal monthly testing point or immediately following a clinical event. Reclassification requires:

- **Upward:** Two consecutive monthly/biweekly training cycles meeting the new group's threshold criteria (for upward movement, e.g., Purple → Gold).
- **Formal:** Formal monthly re-test confirming sustained classification for permanent group reassignment.

Multi-Metric Risk Cluster Patterns

Individual metrics are most powerful when read in combination. The following cluster patterns carry the highest clinical priority and should trigger immediate action regardless of training group assignment:

Risk Cluster	Contributing Metrics	Clinical Priority
Deceleration Deficit	Low EDRFD/BM (<80 N/s/kg) + Low Nordic Max Force (<391 N) + High CM Depth (>34 cm)	HIGHEST — hamstring eccentric and braking force deficit combination; immediate attention regardless of training group
Tendon Compliance	Low LL Stiffness (<6,094 N/m) + Low FV/kg (<23 N/kg) + Low Calf ISO (<1,126 N)	HIGH — Achilles/patellar tendinopathy risk; escalate isometric loading protocol; monitor with daily calf ISO and forceframe tracking
Limb Asymmetry Pattern	CMJ Asymmetry >10% + Nordic Asymmetry >13% same limb + Calf ISO Asymmetry >10% same limb	HIGH — concordant multi-test asymmetry on same limb = structural concern requiring PT evaluation; restrict bilateral high-load work
Low Relative Power	Peak Power/BM <50 W/kg + mRSI <0.50 + Conc. Impulse <254 N·s (Q1)	MODERATE — global capacity deficit; White Group assignment; general strength and power foundation required before phase-specific targeting

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